Hamiltonians and Nonlocal Games

AlgoLab Summer School — Lecture 2

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Prelude: Why Think About Things Other Than Circuits?

Other Quantum Problems

- In Lecture 1 we studied quantum circuits: we asked what useful unitary operations they can implement efficiently.
- **Q:** Why think about any other quantum problems?
- **A:** Several good reasons!
  - **New applications.** Understanding other types of quantum advantage can lead to qualitatively different applications.

    $\rightsquigarrow$ *Example: Quantum Key Distribution (QKD).*
  - **Cross-pollination.** Techniques developed to solve other quantum problems can be ported back to yield quantum computational advantage.

    $\rightsquigarrow$ *We will see an explicit example of this later in the lecture!*
  - **Intuition.** Studying other quantum problems sharpens our intuition for how quantum mechanics behaves.

Part 1: Classical Constraint Satisfaction Problems

Constraint Satisfaction Problems

- Before discussing quantum problems, we introduce an important formalism for modeling classical optimization problems.
- A MAX-CSP is defined by:
  - A set of variables $\{X_1, \ldots, X_n\}$, each taking values in a domain D_i. Today: all variables are binary, $X_i \in \{0, 1\}$.
  - A set of constraints $\{C_1, \ldots, C_m\}$, each a scoring function on a subset of variables:

$$C_i : \{0, 1\}^k \to [0, 1]$$

- Relative to any CSP the value of an assignment $x \in \{0, 1\}^n$ is:

$$\mathrm{val}(x) = \frac{1}{m} \sum_{i=1}^{m} C_i(x)$$

- The problem: find an assignment x^* maximizing $\mathrm{val}(x)$.
- The locality of a CSP is the largest k involved in any constraint. We focus on constant locality: k fixed as n grows.

Example 1: 3XOR

- In a 3XOR instance, each constraint checks the parity of exactly 3 variables:

$$C_{ijk}(X_i, X_j, X_k) = \begin{cases} 1 & \text{if } X_i \oplus X_j \oplus X_k = b_{ijk} \\ 0 & \text{otherwise} \end{cases}$$

  for some target bit $b_{ijk} \in \{0, 1\}$.

- **Example.** Consider the following system of equations over $\mathbb{F}_2$:

$$\begin{aligned} X_1 \oplus X_2 \oplus X_3 &= 0 \\ X_1 \oplus X_2 \oplus X_4 &= 0 \\ X_1 \oplus X_3 \oplus X_4 &= 0 \\ X_2 \oplus X_3 \oplus X_4 &= 1 \end{aligned}$$

- **Q:** Can all four constraints be satisfied simultaneously?
- **Q:** (More general) How hard is it to determine if all constraints in a 3XOR problem can be maximized?
- **A:** Easy! (See next slide.)

Solving 3XOR

- Parity constraints are linear equations over $\mathbb{F}_2$. We can encode the whole system as a matrix equation $Ax = b$:

$$\underbrace{\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{pmatrix}}_{x} = \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{b} \pmod 2$$

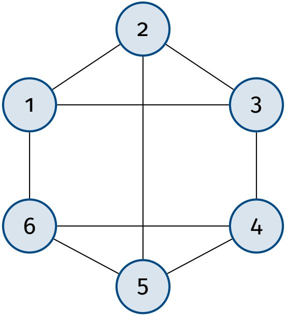
- Then we can check if this system of equations is satisfiable via Gaussian elimination over $\mathbb{F}_2$.
- **Q:** How hard is it to determine the value of a MAX-CSP if all equations cannot be satisfied? (i.e. how hard is it to compute the maximum fraction of satisfiable constraints).
- **A:** Hard! (To be precise: NP-Hard. Will be defined this afternoon!)
- In fact, an even simpler problem is still NP-Hard.

Example 2: Max Cut

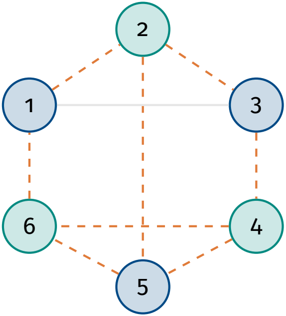
- Given a graph $G = (V, E)$ with $|V| = n$, the Max Cut CSP has:
  - One binary variable $X_i \in \{0, 1\}$ per vertex $i \in V$.
  - One constraint per edge $e = (i, j) \in E$:

$$C_{ij}(X_i, X_j) = \begin{cases} 1 & \text{if } X_i \neq X_j \\ 0 & \text{if } X_i = X_j \end{cases}$$

- Equivalently: partition the vertices into two sets (0-vertices and 1-vertices) to maximise the fraction of edges crossing between them.



Solving Max Cut



8 of 9 edges cut: $\mathrm{val}(x^*) = 8/9$

- **Q:** How hard is Max Cut to solve?
- **A:** Hard! But given a solution, checking how many edges are cut is easy $\Rightarrow$ the problem is in NP.

Aside: Approximation Algorithms

- Random assignments achieve $\mathrm{val} = 1/2$ in expectation.
- Goemans–Williamson: a polynomial-time algorithm achieving at least 0.878 of the optimum value.
- The Unique Games Conjecture implies it is NP-hard to do better.

Part 2: The Local Hamiltonian Problem

Generalizing MAX-CSPs

Local Hamiltonian Problems can be thought of as a quantum generalization of a MAX-CSPs:

- bitstrings $x \in \{0,1\}^n \longrightarrow$ n-qubit states $|\psi\rangle$.
- constraints $C_i : \{0,1\}^k \to [0,1] \longrightarrow$ Hermitian matrices H_i.
- k-local CSP: each C_i involves $\leq k$ bits $\longrightarrow$ k-local Hamiltonian: each H_i acts on $\leq k$ qubits.
- $\max_x \frac{1}{m} \sum_i C_i(x) \longrightarrow \min_{|\psi\rangle} \langle\psi| H |\psi\rangle$.

k-Local Hamiltonian Problem. Given Hermitian matrices $H_1, \ldots, H_m$ each acting nontrivially on at most k qubits of an n-qubit system, find the state $|\psi\rangle$ minimising

$$\langle\psi| H |\psi\rangle, \text{ where } \qquad H = \frac{1}{m}\sum_{i=1}^{m} H_i.$$

Example: Quantum Max Cut

- Given a graph $G = (V, E)$, the Quantum Max Cut (QMC) Hamiltonian is:

$$H_{\mathrm{QMC}} = \frac{1}{|E|} \sum_{(i,j)\in E} H_{ij}, \qquad H_{ij} = \frac{1}{4}\left(I - XX - YY - ZZ\right)_{ij}$$

- Each term H_{ij} acts nontrivially on only qubits i and j — so QMC is a 2-local Hamiltonian problem.
- The goal: find $|\psi\rangle$ minimising $\langle\psi| H_{\mathrm{QMC}} |\psi\rangle$.

Exercise. Show that if we kept only the $Z_i Z_j$ and identity term in the quantum Max Cut Hamiltonian, we would recover the classical Max Cut problem.

Why "Quantum Max Cut"?

Question. What is the ground state (minimum energy state) of the two-qubit QMC Hamiltonian

$$H = \frac{1}{4}(I - XX - YY - ZZ)?$$

And what is the ground state energy?

Why “Quantum Max Cut”?

A: The minimum energy state is the singlet state $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, with energy 0.

Derivation. The key observation is the identity

$$\vec{\sigma}_i \cdot \vec{\sigma}_j := XX + YY + ZZ = 2\,\mathrm{SWAP} - I$$

so:

$$\begin{aligned} H &= \frac{1}{4}(I - XX - YY - ZZ) \\ &= \frac{1}{4}(I - (2\,\mathrm{SWAP} - I)) \\ &= \frac{1}{2}(I - \mathrm{SWAP}) \end{aligned}$$

The eigenvalues of SWAP are $+1$ (symmetric states) and -1 (antisymmetric states), so H has eigenvalues 0 and 1. The ground state energy is **0**, achieved by any antisymmetric (singlet) state $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$.

QMC on More Qubits and Monogamy of Entanglement

Question. Is there a state that achieves energy 0 on the three QMC Hamiltonian associated with a three vertex line?
Equivalently, does there exist a $|\psi\rangle$ on three qubits such that

$$\langle\psi| H_{12} |\psi\rangle = 0 \quad \text{and} \quad \langle\psi| H_{23} |\psi\rangle = 0?$$

A: No! This is an consequence of monogamy of entanglement.

- $H_{12} |\psi\rangle = 0$ forces qubits 1 and 2 to be in the singlet state, which means qubit 2 is maximally entangled with qubit 1.
- But a maximally entangled qubit cannot be entangled with any third system — so qubit 2 cannot simultaneously be in a singlet with qubit 3.

In general, solving QMC is believed to be hard even for a quantum computer. Verifying the energy of a candidate state $|\psi\rangle$ is easy if you have a quantum computer (but hard otherwise). Formally, the problem is QMA-hard (definitions this afternoon!).

References and Further Reading: Quantum Max Cut

- **Problem definition and first approximation algorithm**
  S. Gharibian and O. Parekh, *Almost Optimal Classical Approximation Algorithms for a Quantum Generalization of Max-Cut*, APPROX/RANDOM 2019. arxiv:1909.08846
- **Improved approximation algorithms**
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  Y. Hwang, J. Neeman, O. Parekh, K. Thompson, J. Wright, *Unique Games Hardness of Quantum Max-Cut, and a Conjectured Vector-Valued Borell's Inequality*, SODA 2023. arXiv:2111.01254.
- **Algebraic structure of QMC**
  A. Bene Watts, A. Chowdhury, A. Epperly, J.W. Helton, I. Klep, *Relaxations and Exact Solutions to Quantum Max Cut via the Algebraic Structure of Swap Operators*, *Quantum* 8, 1352 (2024). arXiv:2307.15661.

Part 3: Nonlocal Games

Testing CSPs

- Like the Local Hamiltonian Problem, nonlocal games can be thought of as a quantum generalization of CSPs — but they generalize the way a CSP is tested.
- Imagine a prover (or provers) who already knows the optimal solution to a CSP. We (the verifier) want to be convinced of the solution — but we don't trust the prover. They will always try to convince us the CSP has as high a value as possible.
- **Q:** How could we be convinced of the value of a k-local CSP? For concreteness consider a 3XOR instance like before, but with variables relabeled:

$$A_0 \oplus B_0 \oplus C_0 = 0$$
$$A_1 \oplus B_1 \oplus C_0 = 1$$
$$A_1 \oplus B_0 \oplus C_1 = 1$$
$$A_1 \oplus B_1 \oplus C_0 = 1$$

Testing 3XOR

Q: How could we test the maximum fraction of clauses that can be satisfied in the 3XOR instance:

$$A_0 \oplus B_0 \oplus C_0 = 0$$
$$A_1 \oplus B_1 \oplus C_0 = 1$$
$$A_1 \oplus B_0 \oplus C_1 = 1$$
$$A_1 \oplus B_1 \oplus C_0 = 1$$

One-prover test:

- Ask the prover for a full satisfying assignment; check it.

Three-prover test (Nonlocal game):

- Separate three provers so they cannot communicate. From here on, call them players.
- Pick a clause at random. Send the corresponding 'A' index to player 1, the 'B' index to player 2 and the 'C' index to player 3. (Call these questions). Players respond with a value in $\{0, 1\}$ for the corresponding variable.
- Check if the XOR of the returned values matches the target bit. If so, players win; otherwise they lose.

Soundness of the Three-Prover Test

Claim. The maximum winning probability of the three players playing the nonlocal game equals the value of the 3XOR CSP.

Proof. Let $\tilde{A}_i$ be the response of the first player to question i, then define $\tilde{B}_i$, $\tilde{C}_i$ similarly. These can depend on some randomness λ distributed according to probability distribution $p(\lambda)$. Then the players average win probability is:

$$\frac{1}{4}\sum_{\lambda} p(\lambda)\Big[\Pr\big[\tilde{A}_0(\lambda)+\tilde{B}_0(\lambda)+\tilde{C}_0(\lambda)=0\big]+\ldots \\ +\Pr\big[\tilde{A}_0(\lambda)+\tilde{B}_1(\lambda)+\tilde{C}_1(\lambda)=1\big]\Big]$$

But then picking the value λ which maximizes the expression in the brackets gives a assignment to the CSP variables achieving at least the average win probability of the game. $\square$

Quantum Generalization: The GHZ Game

- Now consider the same three-prover test with one twist: the provers cannot communicate, but they can share entanglement.
- The resulting game is known as the GHZ game.

The GHZ Game.

- The verifier picks a clause uniformly at random from the four clauses of our 3XOR instance.
- Each prover receives the index of their variable (0 or 1) appearing in that clause, and responds with a bit.
- The provers win if their three response bits satisfy the clause.
- Before the game begins, the provers may prepare any shared quantum state. During the game they cannot communicate.

Question. What is the maximum winning probability for the provers in the GHZ game?

Strategy for the GHZ Game

GHZ Strategy. Before the game, the three provers prepare the GHZ state:

$$|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}\big(|000\rangle + |111\rangle\big).$$

During the game each player holds one qubit each. On receiving question $q \in \{0, 1\}$, prover i measures their qubit in one of two bases:

$$q = 0: \ \text{measure in the } X \text{ basis } \{|+\rangle, |-\rangle\}$$
$$q = 1: \ \text{measure in the } Y \text{ basis } \{|+i\rangle, |-i\rangle\}$$

and responds with 0 if the $+$ outcome occurs, 1 if the $-$ outcome occurs.

Claim: this strategy wins the game with probability 1!

Strategy for the GHZ Game

Proof. Write $|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ and note:

$$XXX\,|\mathrm{GHZ}\rangle = XXX\,(|000\rangle + |111\rangle) = (|111\rangle + |000\rangle) = |\mathrm{GHZ}\rangle$$
$$XYY\,|\mathrm{GHZ}\rangle = X(iX)(iX)(IZZ)\,(|000\rangle + |111\rangle) = -\,|\mathrm{GHZ}\rangle$$

(and similarly)

$$YXY\,|\mathrm{GHZ}\rangle = YYX\,|\mathrm{GHZ}\rangle = -\,|\mathrm{GHZ}\rangle$$

so $|\mathrm{GHZ}\rangle$ is a $+1$ eigenstate of *XXX* and a -1 eigenstate of *XYY*, *YXY*, *YYX*.

But this implies that when the players measure *XXX* they will see an even number of $-$ outcomes, and when they measure *XYY*, *YXY*, or *YYX* they will see an odd number $\Rightarrow$ their responses always have the winning parity! $\square$

Part 3: Circuits and Nonlocal Games

Circuit Separations

- Recall from Lecture 1: circuits of 1- and 2-qubit gates can implement any unitary. But physical qubits are subject to noise — the more layers of computation (depth), the more noise accumulates.
- **Q:** Can we achieve quantum advantage with low-depth quantum circuits?
  A: Still unclear!
- **Weaker Q:** Can shallow quantum circuits achieve quantum advantage over shallow *classical* circuits?
  A: Yes!

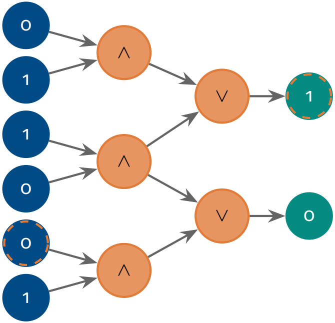
Theorem (Bravyi–Gosset–König, 2018). There exists a computational problem solvable by a constant-depth quantum circuit that cannot be solved by any constant-depth classical circuit.

Quantum Advantage with Shallow Circuits 1

Theorem. QNC^0 circuits (constant-depth, 1- and 2-qubit gates) can solve a relational problem (mapping input bits to output bits) that NC^0 circuits (constant-depth, unbounded fan-out, AND/OR gates) cannot.

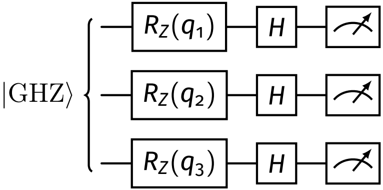
Proof sketch, Part 1: Classical lower bound.

- In an NC^0 circuit, each output bit depends on only a constant number of input bits.
- So an NC^0 circuit cannot win the GHZ game: some player's output is independent of another player's question, breaking the required correlations.



Proof sketch, Part 2: Quantum upper bound.

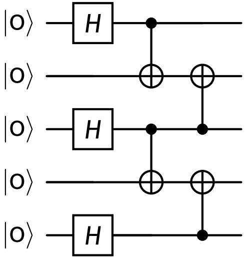
- Players share a GHZ state. Each receives a question bit q_i, applies a classically-controlled R_Z rotation, then measures in the X basis — implementing the GHZ strategy in constant depth.



Problem. A GHZ state cannot spanning n all qubits cannot be prepared by a QNC^0 circuit — its preparation requires depth at least $\Omega(\log n)$.

Proof sketch, Part 3: Fix with cluster states.

- A cluster state sharing GHZ-like entanglement can be prepared in constant depth.



- Preparing this state and making GHZ measurements controlled on inputs produces non-classical correlations. □

References and Further Reading: Circuit Separations

- **Original circuit separation**
  S. Bravyi, D. Gosset, R. König, *Quantum Advantage with Shallow Circuits*, Science 362, 308–311 (2018). arXiv:1704.00690.
- **Improvement: separation against unbounded fan-in classical circuits**
  A. Bene Watts, R. Kothari, L. Schaeffer, A. Tal, *Exponential Separation Between Shallow Quantum Circuits and Unbounded Fan-In Shallow Classical Circuits*, STOC 2019. arXiv:1906.08890.
- **Separation with noise**
  S. Bravyi, D. Gosset, R. König, M. Tomamichel, *Quantum Advantage with Noisy Shallow Circuits*, Nature Physics 16, 1040–1045 (2020). arXiv:1904.01502.
- **Experimental implementation (trapped-ion)**
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